

4. Kernel aur uske Core Operations (With Types of Kernel)

Computer Science Fundamentals | Module 1: OS Architecture | CS_Fundamental-00

Reference to Previous Concepts: Concept 1.1 mein humne dekha tha ki OS hardware manage karta hai aur Concept 3.2 (System View) mein Kernel ko core component bataya gaya tha. Yeh topic **Image 4** par aadharit hai jo Kernel ke **5 Fundamental Operations** aur **3 Types of Kernel** ko detail karta hai.

4.1 Kernel kya hota hai (The Core Bridge)

- **Heart of the OS:** Kernel OS ka core program hai jo boot hote hi RAM mein load hota hai aur system chalne tak active rehta hai.
- **Bridge Role:** Yeh hardware aur software ke beech direct bridge hai. Applications hardware ko direct touch nahi kar sakte; unhe Kernel se **System Calls** ke through permission leni padti hai.

4.2 Key Functions / Operations of Kernel (5 Pillars)

1. **Process Management:** CPU time allot karna (CPU Scheduling), process creation, coordination aur termination ensure karna.
2. **Memory Management:** RAM allocation/deallocation, Memory Protection (isolation), aur Virtual Memory handle karna.
3. **Device Management:** Hardware peripherals ke device drivers handle karna aur uniform I/O interface provide karna.
4. **File System Management:** Storage devices (SSD/HDD) par files create, read, write aur organize karna.
5. **Security & Access Control:** User permissions verify karna, unauthorized access rokna aur system integrity maintain karna.

Fig 4.1: Kernel Architecture & Core Operations

Reference: Image 4 [Kernel Functions & 3 Types Comparison]

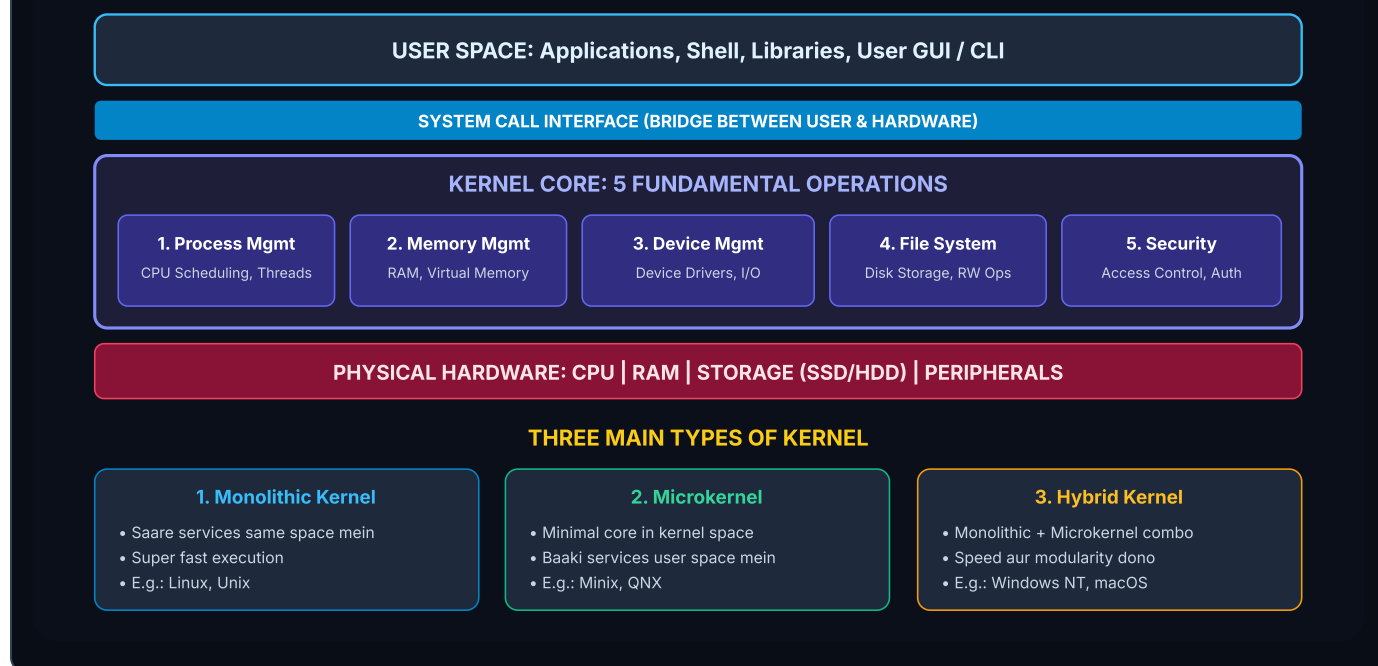


Fig 4.1: Kernel Architecture, 5 Operations & 3 Kernel Types

4.3 Types of Kernel (Image 4 Reference)

4.3.1 Monolithic Kernel

- Definition:** Saare OS services (Process, Memory, File System, Drivers) **ek hi address space (Kernel Space)** mein run karte hain.
- Pros:** Super fast execution (direct memory function calls).
- Cons:** Agar ek driver crash ho, toh poora OS crash ho sakta hai.
- Examples:** Linux, Traditional Unix.

4.3.2 Microkernel

- Definition:** Kernel space ko **bahut chhota (minimal)** rakhta hai. Baaki services User Space mein servers ke roop mein chalti hain.
- Pros:** Highly secure aur crash-resistant (Driver crash hone se OS nahi marta).
- Cons:** IPC (message passing) overhead ki wajah se thoda slow.
- Examples:** Minix, QNX, L4.

4.3.3 Hybrid Kernel

- Definition:** Monolithic ki high speed aur Microkernel ki modularity ko combine karta hai.
- Examples:** Windows NT (Windows 10/11), macOS (XNU).

4.4 Kernel Types Comparison Table

Paimana	Monolithic Kernel	Microkernel	Hybrid Kernel
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Address Space	Saare services Kernel Space mein	Sirf core kernel space mein, baaki User Space	Mixture (Critical parts in kernel space)
Speed	Bahut Fast	Thoda Slow (IPC overhead)	Fast & Balanced
Fault Tolerance	Low (Driver crash = OS crash)	Very High (Stable)	High
Real Examples	Linux, Unix	Minix, QNX	Windows NT, macOS